

PREPRINT: Estimating Infection Severity for SPOK-24 from Early Household and Hospital Data in Australia

Version 1: January 7, 2025.

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Summary

Background

SPOK-24, a newly identified zoonotic paramyxovirus, emerged in Brisbane, Australia in September 2024. To assess its early clinical severity, we estimated the infection fatality ratio (IFR) and infection hospitalization ratio (IHR) among symptomatic infections.

Methods

We analyzed data from symptomatic cases drawn from: (1) a prospective cohort of 247 Brisbane-area households with laboratory-confirmed SPOKARS-24 infections, (2) clinical admissions across six Queensland hospitals, and (3) diagnostic surveillance of repatriation flights from Australia between Oct 14 and December 15, 2024. Diagnosis was confirmed via RT-PCR targeting the SPOK-24 L gene. Asymptomatic infection was estimated at 20% (95% CI, 12–31), based on serologic follow-up and contact testing.

Findings

The estimated IFR among symptomatic individuals was 0.03% (95% CI 0.004–0.11) for individuals aged 0–17 years, 0.06% (0.02–0.16) for those 18–49 years, 0.1% (0.05–0.26) for 50–64 years, and 4.0% (1.8–8.4) for those 65 years and older. The IHR for the same age groups was 1.9% (0.4–4.8), 0.5% (0.2–1.7), 1.5% (0.4–2.9), and 20.1% (9.4–32.1), respectively. Pediatric fatalities were observed in both household and hospital cohorts, confirming risk in younger populations.

Interpretation

While overall severity of SPOK-24 appears lower than that of classical Spok viruses, the high hospitalization rates among older adults and observed pediatric deaths underline the need

for close clinical monitoring. Our findings support the ongoing classification of SPOK-24 as a pathogen of high pandemic potential.

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